

Basic CSS — 2-Hour Teaching Notes

Total time: 2 hours (120 minutes) **Audience:** Beginners with basic HTML knowledge **Goal:** Understand what CSS is, how to apply it, and use core properties to style a webpage confidently.

Session Outline

Time	Topic
0:00-0:10	Introduction to CSS
0:10-0:25	Ways to add CSS
0:25-0:45	Selectors
0:45-1:00	Colors, Units & Text Properties
1:00-1:10	Break
1:10-1:35	Box Model
1:35-1:55	Display, Positioning & Backgrounds
1:55-2:00	Wrap-up, Q&A, Practice Task

1. Introduction to CSS (10 min)

- **CSS** = Cascading Style Sheets
- Controls **presentation/style** of HTML: colors, fonts, spacing, layout
- HTML = structure (skeleton), CSS = appearance (skin/clothes)
- “Cascading” = rules can override each other based on specificity, order, and source

Analogy: HTML is a house’s rooms and walls; CSS is the paint, furniture, and decoration.

2. Ways to Add CSS (15 min)

a) Inline CSS

```
<p style="color: red;">Hello</p>
```

- Applied directly on an element
- Highest priority but hard to maintain — avoid for large projects

b) Internal CSS

```
<head>  
  <style>  
    p { color: blue; }  
  </style>  
</head>
```

- Written inside <style> tag in the HTML file
- Good for single-page demos

c) External CSS (best practice)

```
<link rel="stylesheet" href="style.css">

/* style.css */
p { color: green; }
```

- Kept in a separate .css file
- Reusable across multiple pages, easiest to maintain

Teaching tip: Demo all three live, show how order/specificity changes which style “wins.”

3. Selectors (20 min)

Selector	Example	Selects
Universal	* { }	All elements
Element	p { }	All <p> tags
Class	.box { }	Elements with class="box"
ID	#header { }	Element with id="header"
Grouping	h1, h2, p { }	All listed elements
Descendant	div p { }	<p> inside a <div>
Child	div > p { }	Direct <p> children of <div>
Pseudo-class	a:hover { }	Link on hover

Class vs ID: - Class → reusable, multiple elements (.) - ID → unique, one element per page (#)

Practice: Give students a small HTML snippet, ask them to write selectors targeting specific elements.

4. Colors, Units & Text Properties (15 min)

Colors

```
color: red; /* named color */
color: #ff0000; /* hex */
color: rgb(255, 0, 0); /* rgb */
color: rgba(255,0,0,0.5); /* rgb with opacity */
```

Units

- **Absolute:** px
- **Relative:** %, em (relative to parent font size), rem (relative to root font size)

Text Properties

```
font-family: Arial, sans-serif;
font-size: 16px;
font-weight: bold;
text-align: center;
text-decoration: underline;
line-height: 1.5;
```

☕ Break (10 min)

5. Box Model (25 min)

Every HTML element is a rectangular box made of:

```
[ margin
  [ border
    [ padding
      [ content ]
    ]
  ]
]
```

- **Content** – actual text/image
- **Padding** – space between content and border
- **Border** – edge around padding
- **Margin** – space outside the border, between elements

```
div {
  width: 200px;
  padding: 20px;
  border: 2px solid black;
  margin: 10px;
}
```

box-sizing

```
* { box-sizing: border-box; }
```

- content-box (default): width/height apply to content only
- border-box: width/height include padding + border (easier to manage — recommend by default)

Demo: Use browser DevTools to visually inspect the box model on a live element.

6. Display, Positioning & Backgrounds (20 min)

Display

```
display: block;      /* full width, new line (div, p) */
display: inline;     /* no new line, no width/height (span, a) */
display: inline-block; /* inline but accepts width/height */
display: none;       /* hides element completely */
```

Positioning (brief intro)

```
position: static;    /* default */
position: relative;  /* offset from normal position */
position: absolute;  /* positioned relative to nearest positioned
ancestor */
position: fixed;     /* stays fixed on screen while scrolling */
```

Backgrounds

```
background-color: lightblue;
```

```
background-image: url("bg.jpg");
background-size: cover;
```

7. Wrap-up, Q&A & Practice Task (5 min)

Mini practice task (assign as homework or last-5-min activity):

> Style a simple profile card: > - Box with border, padding, and rounded corners (border-radius) > - Centered text > - Background color > - Image with fixed width

Recap key takeaways: 1. CSS = styling layer for HTML 2. Three ways to add CSS — external is best practice 3. Selectors target elements to style 4. Box model governs spacing of every element 5. Display & position control layout behavior

Suggested Homework / Next Session Preview

- Practice building a styled card component
 - Next session: **Flexbox & Grid** for layouts, and **responsive design** basics
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8. Mini Project: Student Registration Form (Hands-on)

A small practice project that uses **only the CSS covered above** — selectors, colors, units, text properties, box model, display, position, and backgrounds. No Flexbox/Grid used.

HTML (index.html)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Student Registration Form</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="form-container">
    <h2 class="form-title">Student Registration Form</h2>

    <form>
      <div class="form-group">
        <label for="name">Full Name</label>
        <input type="text" id="name" placeholder="Enter your full
name">
      </div>

      <div class="form-group">
        <label for="email">Email Address</label>
        <input type="email" id="email" placeholder="Enter your
email">
      </div>

      <div class="form-group">
        <label for="course">Course</label>
```

```

        <select id="course">
            <option>Web Development</option>
            <option>Data Science</option>
            <option>Graphic Design</option>
        </select>
    </div>

    <div class="form-group">
        <label for="dob">Date of Birth</label>
        <input type="date" id="dob">
    </div>

    <button type="submit" class="submit-btn">Register</button>
</form>
</div>

</body>
</html>

```

CSS (style.css)

```

/* Universal selector - reset spacing */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

/* Element selector for body */
body {
    font-family: Arial, sans-serif;
    background-color: #f0f4f8;
    display: block;
    padding: 40px 0;
}

/* Class selector for the card */
.form-container {
    width: 400px;
    margin: 0 auto;           /* center the box horizontally */
    background-color: #ffffff;
    padding: 30px;
    border: 1px solid #ccc;
    border-radius: 10px;
    position: relative;
}

/* Class selector for heading */
.form-title {
    text-align: center;
    color: #2c3e50;
    font-size: 22px;
    margin-bottom: 20px;
}

/* Descendant selector - labels inside form-group */
.form-group {
    margin-bottom: 15px;
}

.form-group label {
    display: block;           /* label sits on its own line */
    font-size: 14px;
    font-weight: bold;
    color: #34495e;
}

```

```

    margin-bottom: 5px;
}

/* Descendant selector - inputs/select inside form-group */
.form-group input,
.form-group select {
    width: 100%;
    padding: 10px;
    font-size: 14px;
    border: 1px solid #bbb;
    border-radius: 5px;
}

/* Pseudo-class - highlight input on focus */
.form-group input:focus,
.form-group select:focus {
    border: 1px solid #3498db;
    background-color: #f7fbff;
}

/* Class selector for button */
.submit-btn {
    display: block;
    width: 100%;
    padding: 12px;
    background-color: #3498db;
    color: white;
    font-size: 16px;
    font-weight: bold;
    text-align: center;
    border: none;
    border-radius: 5px;
}

/* Pseudo-class - button hover effect */
.submit-btn:hover {
    background-color: #2980b9;
}

```

CSS Concepts Used in This Project

Concept	Where it's used
Universal selector *	Resetting margin/padding, box-sizing
Element selector	body { }
Class selector	.form-container, .form-title, .submit-btn
Descendant selector	.form-group label, .form-group input
Pseudo-class	:focus, :hover
Colors	Named, hex (#2c3e50, #3498db)
Units	px, %
Text properties	font-family, font-size, font-weight, text-align
Box model	padding, margin, border, border-radius, box-sizing: border-box
Display	display: block on labels and button
Positioning	position: relative on the container
Background	background-color on body, container, and button

Classroom activity idea: 1. Have students type out the HTML first and view it unstyled in the browser. 2. Then add the CSS step by step (structure → box model → colors/text → hover/focus states), asking

students to observe the visual change after each block. 3. End with a short challenge: change the theme color, add a "Phone Number" field, or round the corners differently.